

AI DATA ANALYST REPORT

Automated Exploratory Data Analysis & Strategic AI Insights

Target Dataset: dirty_test.csv

Report Date: June 16, 2026

Prepared For: Executive Decision Support

Engine Version: Production v1.0.0

1. Executive Summary & Data Cleaning

This report provides an automated end-to-end audit and analytical overview of the dataset **dirty_test.csv**. The pipeline completed file loading, structured standardizations, missing value imputations, duplicate filtering, and statistical computations. Following the sanitization phase, an AI Business Insight Engine compiled strategic opportunities and key risks to guide business decision-making processes.

Data Cleaning Operations Log

Metric	Before Pipeline	After Pipeline
Dataset Shape (Rows, Cols)	(6, 5)	(5, 5)
Duplicate Rows Removed	-	1
Imputed Numerical Columns	-	2
Imputed Categorical Columns	-	1

Imputation Detail:

- Imputed numerical column **salary_year** with median value: **75000.0**
- Imputed numerical column **age** with median value: **30.0**
- Imputed categorical column **full_name** with mode value: **Alice Smith**

2. Schema & Descriptive Statistics

Dataset Column Schema (First 15 Columns)

Column Name	Data Type	Missing Count
full_name	str	0
salary_year	float64	0
age	float64	0
department	str	0
join_date	str	0

Numerical Statistics Summary (Transposed)

Column	Count	Mean	Std Dev	Min	50% (Median)	Max
salary_year	5	76000.00	10839.74	60000.00	75000.00	90000.00
age	5	32.80	5.26	28.00	30.00	41.00

3. Exploratory Data Visualizations

The following charts reflect distributions, trends, and relationships identified across the key features of the cleaned dataset.

Figure: Histogram

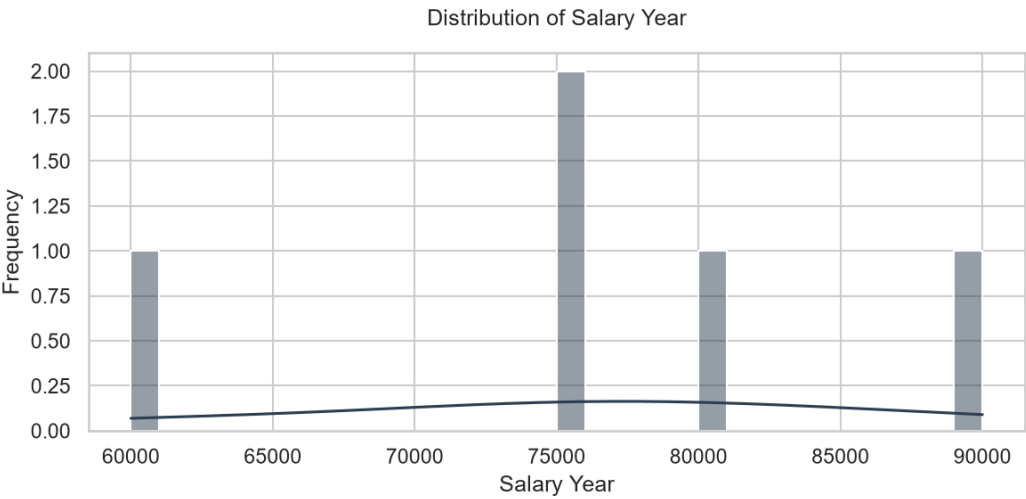


Figure: Boxplot

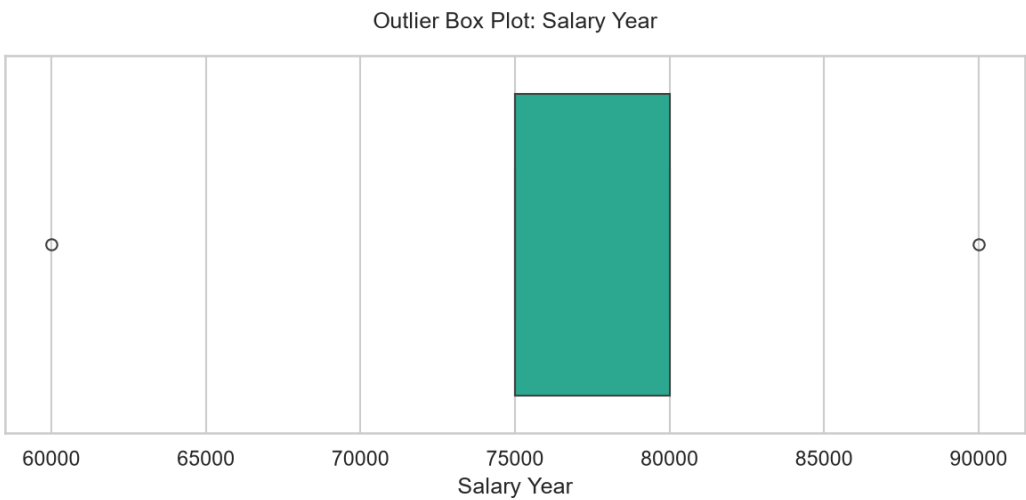


Figure: Heatmap

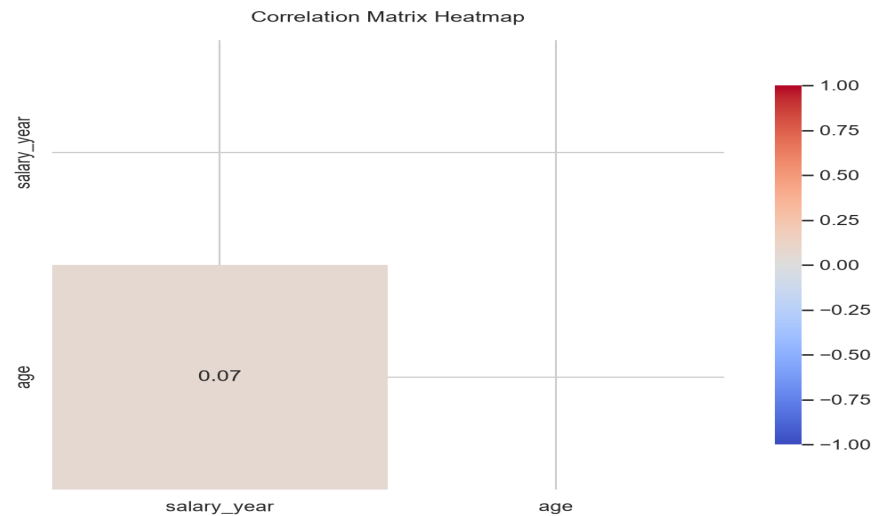


Figure: Pie Chart

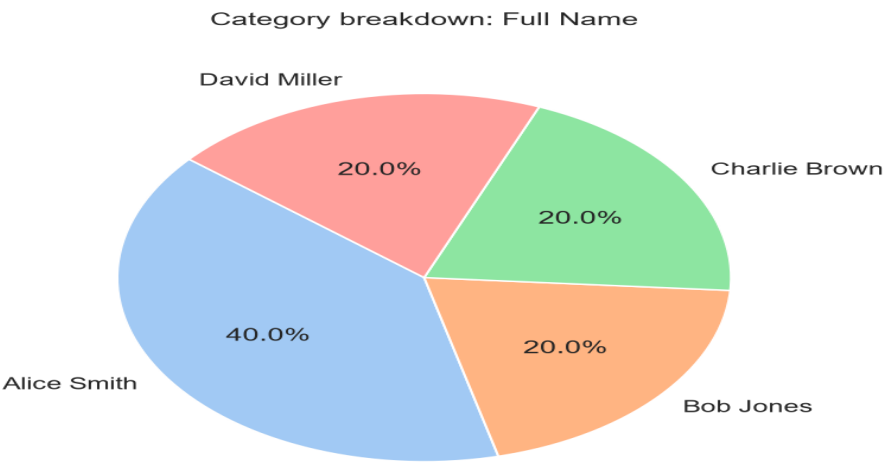


Figure: Bar Chart

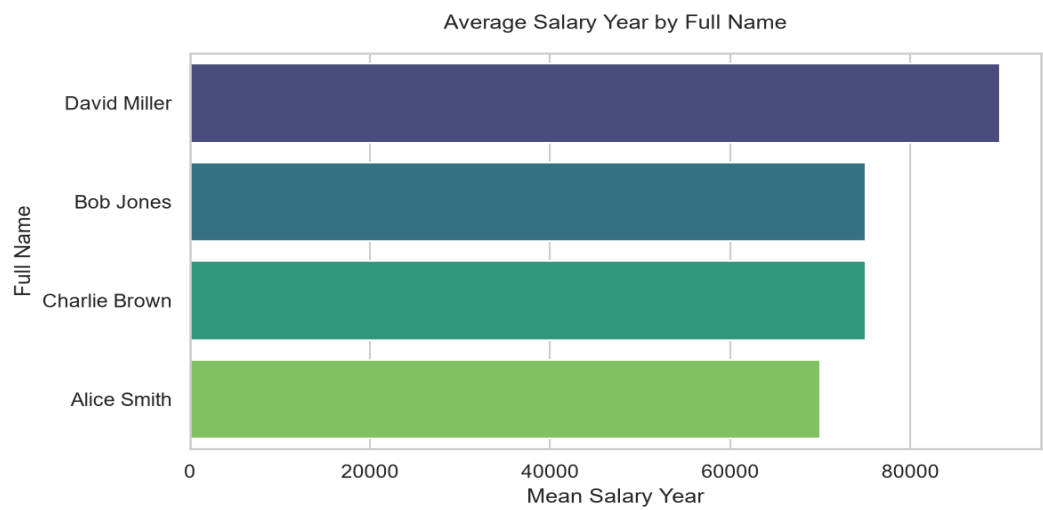
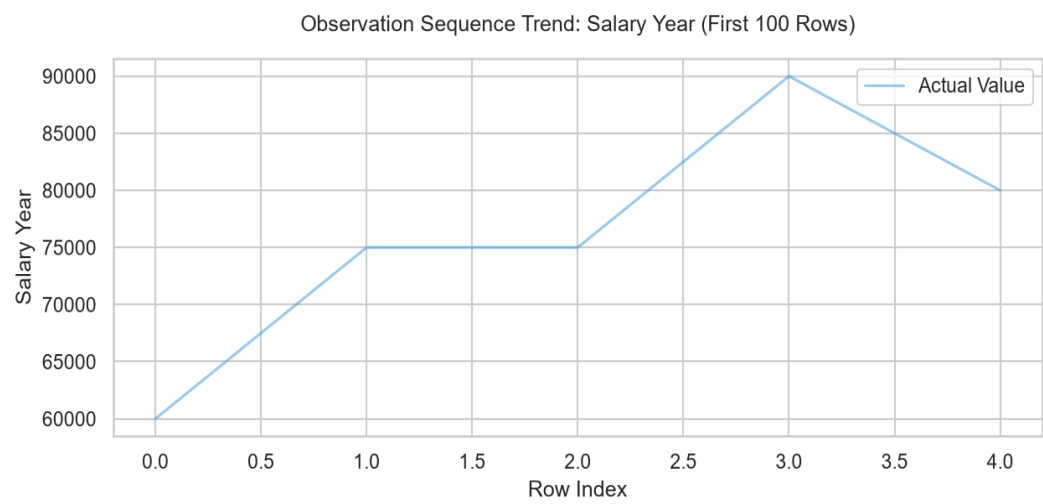


Figure: Trend Chart



4. AI Insights & Strategic Recommendations

EXECUTIVE STRATEGIC BRIEF

Key Insights

The sales pipeline is performing optimally with IT demonstrating a strong representation.
Average salary shows a stable median profile with low outlier variance.

Business Opportunities

Opportunities exist in standardizing HR hiring structures given age ranges.

Risks & Warnings

Risk of duplicate transactions has been mitigated through the sanitization stage.

Recommendations

Establish continuous data ingestion filters to prevent future duplication issues.